

03 - Doubly Linked Lists

Due Sep 29 at 2pm **Points** 1 **Questions** 7 **Available** until Sep 29 at 2pm
Time Limit 30 Minutes **Allowed Attempts** 10

This quiz was locked Sep 29 at 2pm.

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	23 minutes	0.95 out of 1

❗ Correct answers are hidden.

Score for this attempt: **0.95** out of 1

Submitted Sep 25 at 6:58pm

This attempt took 23 minutes.

Question 1	0.06 / 0.06 pts
What does each method do?	
precede(q,r:E2*)	Inserts (*q) before (*r) ⇅
follow(p,q:E2*)	Inserts (*q) after (*p) ⇅
unlink(q:E2*)	Removes (*q) from the list ⇅

Question 2	0.05 / 0.05 pts

Which operation of C2Ls is shown in the image?

$p := q \rightarrow prev ; r := q \rightarrow next$
$p \rightarrow next := r ; r \rightarrow prev := p$
$q \rightarrow prev := q \rightarrow next := \text{this}$

- ☒ unlink($q:E2^*$)
- ☐ follow($p,q:E2^*$)
- ☐ precede($q,r:E2^*$)

Question 3

0.05 / 0.05 pts

Which operation of C2Ls is shown in the image?

$r := p \rightarrow next$
$q \rightarrow prev := p ; q \rightarrow next := r$
$p \rightarrow next := r \rightarrow prev := q$

- ☒ follow($p,q:E2^*$)
- ☐ unlink($q:E2^*$)
- ☐ precede($q,r:E2^*$)

Question 4

0.05 / 0.05 pts

Which operation of C2Ls is shown in the image?

$p := r \rightarrow prev$
$q \rightarrow prev := p ; q \rightarrow next := r$
$p \rightarrow next := r \rightarrow prev := q$

☐ unlink($q:E2^*$)

☒ precede($q,r:E2^*$)

☐ follow($p,q:E2^*$)

Question 5

0.4 / 0.4 pts

Complete the following pseudocode so that the algorithm solves the following task:

Create a method that counts how many times an element with a key k appears in a C2L list. While counting, also remove the elements with key k .

• #:

SearchAndDelete($\#L:E2^*, k:N$) :N

p:= @	
c:=0	
d	
p->key=k	
c:=c+1	
ß	L
§	
Ð	
\$	
return c	

[Select]



- @: [Select]
- đ: [Select]
- ß: [Select]
- §: [Select]
- Đ: [Select]
- \$: p:=r
- Ł: [Select]

Answer 1:

(empty)

Answer 2:

p:=L->next

Answer 3:

p≠L

Answer 4:

r:=p->next

Answer 5:

unlink(p)

Answer 6:

delete(p)

Answer 7:

p:=r

Answer 8:

p:=p->next

Question 6

0.09 / 0.09 pts

L points to a C2L's (cyclic, two-way list) header. Mark the true statements.

☐ If the list is empty then $L = \emptyset$



To insert a new element in the list, it is sufficient to use only two pointers: one of them points to the element to be inserted, the other one points to the list element before/next to which the new element will fit.

☐ The $L \rightarrow \text{next} \rightarrow \text{next}$ expression throws an error if L is an empty list.

☒ To get the penultimate element of the list, $\Theta(1)$ steps are needed.



For every element in the list, its *next* and *prev* pointers cannot be $\emptyset$ (the null pointer)

Partial

Question 7

0.25 / 0.3 pts

A cyclic two-way list is given. The pointer L points to the header of this list, which contains integers and is unordered. While traversing the list once, we delete the smaller elements than their left neighbour. The resulting list will be nondecreasing.

Select the missing part of the pseudo code from the list!

DeletingSort(L:E2*)

p:=L->next

[Select]



[Select]



if (p->key < p->prev->key)

s := p



```
else
    p := p->next
end if
end
```

Answer 1:

p:=L->next

Answer 2:

while

Answer 3:

p ≠ L

Answer 4:

p:=p->next

Answer 5:

unlink(s)

Answer 6:

delete s

Quiz Score: **0.95** out of 1